

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Theory of Computation Course Code: CSPC-512 Examination: 1st Periodical Examination

Scheme: NEP Branch: CSE (AI & ML) Semester: V

Question Paper for 1st Periodical Examination (September 2025) (Session July-November 2025)

Time: One and a half hours

Max. Marks: 30

Note: Attempt all questions.

Course Outcomes:

CO1. Understand the fundamental concepts of automata theory and finite state machines, and analyze, design, and apply deterministic and non-deterministic finite automata, including equivalence and minimization techniques.

CO2. Understand regular languages and expressions, closure properties, Arden's theorem, Pumping Lemma, and the Myhill-Nerode theorem, and apply them for formal language analysis.

CO3. Understand Chomsky Formalism, including regular and context-free grammars, their structure, simplification, standard forms, and the classification of languages in the Chomsky hierarchy.

CO4. Study pushdown automata as computational models for context-free languages and apply them to relevant problems.

CO5. Explore Turing machines and apply concepts of decidability to problems such as the halting problem and post correspondence problem.

Q.1 Answer the following questions:

(5)

- (a) Define a finite automaton.
- (b) State any 2 closure properties of regular set.
- (c) What is an epsilon-transition in NFA?
- (d) State Myhill-Nerode theorem.

Q.2 Explain Pumping Lemma for regular languages with an example.

(5)

Q.3 Prove that the language $\{a^n b^n \mid n \geq 0\}$ is not regular using the Pumping Lemma.

(5)

Q.4 State Arden's Theorem and illustrate it with a simple example of a regular expression.

(5)

Q.5 Compare Moore and Mealy machines in terms of state output, transition output, and timing of output. Illustrate with a simple example for each type.

(5)

Q.6 Explain the steps to convert the given NFA into DFA.

(10)

